Prepare DP1 paper (SP-2021)

[SP-2833] Respond to review feedback on Section 4: Data Processing

Created: 05/Dec/25 Updated: 12/Feb/26 Resolved: 12/Feb/26

Status:	Done
Project:	System Performance
Components:	None
Fix versions:	None
Parent:	Prepare DP1 paper

Type:	Story	**Priority:**	Undefined
Reporter:	Leanne Guy	**Assignee:**	Leanne Guy
Resolution:	Done	**Votes:**	0
Labels:	None		
Remaining Estimate:	Not Specified		
Time Spent:	Not Specified		
Original estimate:	Not Specified		

Reviewers:	James Mullaney
Sprint:	

Description

Section 4:

Section 4.6.1 - I presume the column name for the machine learning reliability (real/bogus score) is the reliability column. This value is between 0 and 1?

Section 4.6.3 - I don't really follow the discussion about heliolinc runs testing out to 9.8 au. There are things beyond 10 au in the Solar System? Was the dataset not sensitive to them?

Section 4.6.3 - I would expect some sort of orbit table tallying the number of objects in different types of orbits or discussion of what was found for moving Solar System. Were all the discoveries Main Belt Asteroids? As there is a Minor Planet Center orbit table/catalog this information must be available. Please add some summary statistics and figures about the Solar System objects associated and new objects found in DP1.

Section 4.6.3 - I don't understand what associating a DiaSource to a known Solar System object means. Does that mean those DiaSource observations are not used to look for new moving objects as it is deemed to be a known moving object. Does this run separately from heliolinx? 1 arcsecond seems a big difference, were there any misidentifications?

Section 4.6.3 - Various papers mention that there must be 3 tracklets over 15 day requirement for the Rubin pipelines to identify/discover a new Solar System object in LSST data. Do these requirements apply for DP1 as well? Do sources need to have moved between the nightly observations or can a tracklet be made of a transient source that hasn't moved in a single night but not visible on another night?
Are there an upper and lower limits on the on-sky angular velocity that can be linked using heliolinx?

Section 4.6.3 There's mention of heliolinx and also heliolinc runs - Can you clarify the differences between these? Is one the algorithm and the other a wrapper around it?

Section 4.6.3 - "link_purify provided a final, non-overlapping set of high-confidence asteroid candidates, ranked by orbit-fit residuals and related metrics" I don't understand the discussion about "link_purify" and how that distills to the SSObject/SSSource table. Some clarifying text would be helpful here. Were there false positive linkages? How many new discoveries versus detections of previous known objects? Moving or replicating the first part of Section 5.10.1 to this section would be helpful.

Figure 36 - "The wider scatter in the RA direction is due to objects whose measured orbital elements are less well constrained, translating to larger along-track positional errors in the predicted positions" - Why not only evaluate Solar System objects with secure orbits/low positional uncertainty?

Guidance for the DP1 Users:

I see from the data schema that there are several aperture fluxes in all the catalogs containing photometry. It would be helpful to have a table similar to Table 6 in Abbott et al (2018; https://ui.adsabs.harvard.edu/abs/2018ApJS..239...18A/abstract) that gives the aperture diameter/radius in pixels and arcseconds.

I think there needs to be more guidance to the reader of what to use for doing their science. For example in the diaObject catalog there is psfFlux, scienceFlux, trailFlux, apFlux, forced_PsfFlux, aperture fluxes. In the object catalog there are by my count ~16 fluxes per filter, and there are 2 magnitudes per filter for objects. What should someone be using for photometry of an extended elliptical galaxy for photometric redshifts, or a stellar-PSF source in the object catalog? What flux values should be used for a supernovae or a trailed moving Solar System object in the diaObject table? Some guidance in this paper I thnk would be useful given the impact DP1 will have as the Rubin/LSST training set.

Why is there no trailFluxErr, trailRAErr, and trailDecErr in the diaObject table? Is there a reason these values were no generated? For those science cases that would need these values, what are the recommendations to use instead within the DP1 data schema?

I couldn't find anywhere in the paper that mentioned that the fluxes are measured/reported in nJy in the stored catalogs. Please add this information to the paper or make it clearer/more obvious for the reader if this information is already available in the paper.
Similar data release papers describe/highlight the most useful set of pixel flags and parameters stored in the database that user should be reviewing when interrogating the released dataset. Having a section/table that presents that information for DP1 would significantly enhance the manuscript.

The appears to be overlap between the Dark Energy Survey footprint and the ComCam observing fields, but there's no filter transforms reported in this paper. This may be out of scope for this paper but as there is an draft technote online (https://rtn-099.lsst.io /) , I strongly encourage the authors to consider finalizing that analysis and including it in this paper and as it would significantly extend the utility of the DP1 dataset.

Line 2269 - Applying a reliability threshold improves the purity transients but not variable stars. What reliability value is recommended for the user looking for transients. A figure may be useful to enable readers to make an informed decision on what threshold is best for their science case.

Comments

Comment by Leanne Guy [27/Jan/26]

Section 4.6.1 - I presume the column name for the machine learning reliability (real/bogus score) is the reliability column. This value is between 0 and 1?

> yes added text

Section 4.6.3 - I don't really follow the discussion about heliolinc runs testing out to 9.8 au. There are things beyond 10 au in the Solar System? Was the dataset not sensitive to them?

> Ari thinks the heliolinc run being described stopped at 10 AU because you'd want to use different parameters for a run targeting more distant objects.
>
> Sentence added to text to clarify

Section 4.6.3 - I would expect some sort of orbit table tallying the number of objects in different types of orbits or discussion of what was found for moving Solar System. Were all the discoveries Main Belt Asteroids? As there is a Minor Planet Center orbit table/catalog this information must be available. Please add some summary statistics and figures about the Solar System objects associated and new objects found in DP1.

> 4.6.3 is the DRP section and describes the processing. This is a result and would go in section 5.9 : Solar System ….. Response will be addeed on https://rubinobs.atlassian.net/browse/SP-2834

Section 4.6.3 - I don't understand what associating a DiaSource to a known Solar System object means. Does that mean those DiaSource observations are not used to look for new moving objects as it is deemed to be a known moving object. Does this run separately from heliolinx? 1 arcsecond seems a big difference, were there any misidentifications?

> Description added to 4.6.3

Section 4.6.3 - Various papers mention that there must be 3 tracklets over 15 day requirement for the Rubin pipelines to identify/discover a new Solar System object in LSST data. Do these requirements apply for DP1 as well? Do sources need to have moved between the nightly observations or can a tracklet be made of a transient source that hasn't moved in a single night but not visible on another night?
Are there an upper and lower limits on the on-sky angular velocity that can be linked using heliolinx?

> That is right, the requirements do apply; to claim a discovery candidate, we required tracklets to be linked over at least three nights.
> Sentence added to section

Section 4.6.3 There's mention of heliolinx and also heliolinc runs - Can you clarify the differences between these? Is one the algorithm and the other a wrapper around it?

> rewrote section to clarify

Section 4.6.3 - "link_purify provided a final, non-overlapping set of high-confidence asteroid candidates, ranked by orbit-fit residuals and related metrics" I don't understand the discussion about "link_purify" and how that distills to the SSObject/SSSource table. Some clarifying text would be helpful here. Were there false positive linkages? How many new discoveries versus detections of previous known objects? Moving or replicating the first part of Section 5.10.1 to this section would be helpful.

> S 5.10.1 describes performance and 4 the algorithm, so I will keep them separate.
>
> Added clarification to linking step

Figure 36 (now 38) - "The wider scatter in the RA direction is due to objects whose measured orbital elements are less well constrained, translating to larger along-track positional errors in the predicted positions" - Why not only evaluate Solar System objects with secure orbits/low positional uncertainty?

> Add comment in astrometric residuals for asteroid association

I see from the data schema that there are several aperture fluxes in all the catalogs containing photometry. It would be helpful to have a table similar to Table 6 in Abbott et al (2018; https://ui.adsabs.harvard.edu/abs/2018ApJS..239...18A/abstract) that gives the aperture diameter/radius in pixels and arcseconds.

> This information is in the https://sdm-schemas.lsst.io/dp1.html - I will reference it in the DP1 paper but not make a new table.

I think there needs to be more guidance to the reader of what to use for doing their science. For example in the diaObject catalog there is psfFlux, scienceFlux, trailFlux, apFlux, forced_PsfFlux, aperture fluxes. In the object catalog there are by my count ~16 fluxes per filter, and there are 2

magnitudes per filter for objects. What should someone be using for photometry of an extended elliptical galaxy for photometric redshifts, or a stellar-PSF source in the object catalog? What flux values should be used for a supernovae or a trailed moving Solar System object in the diaObject table? Some guidance in this paper I thnk would be useful given the impact DP1 will have as the Rubin/LSST training set.

> This is a good topic for a DP1 tutorial. I will include this information in future DP/DR papers.

Why is there no trailFluxErr, trailRAErr, and trailDecErr in the diaObject table? Is there a reason these values were no generated? For those science cases that would need these values, what are the recommendations to use instead within the DP1 data schema?

> There is no trailFlux, etc in the diaObject table, so no Err either. They are in the DiaSource table but no associated errors.

I couldn't find anywhere in the paper that mentioned that the fluxes are measured/reported in nJy in the stored catalogs. Please add this information to the paper or make it clearer/more obvious for the reader if this information is already available in the paper.

> Added in data_products → catalogs

Similar data release papers describe/highlight the most useful set of pixel flags and parameters stored in the database that user should be reviewing when interrogating the released dataset. Having a section/table that presents that information for DP1 would significantly enhance the manuscript.

> I'm not going to add this for DP1. It will be done in future data releases, and perhaps for DP2

The appears to be overlap between the Dark Energy Survey footprint and the ComCam observing fields, but there's no filter transforms reported in this paper. This may be out of scope for this paper but as there is an draft technote online (https://rtn-099.lsst.io /) , I strongly encourage the authors to consider finalizing that analysis and including it in this paper and as it would significantly extend the utility of the DP1 dataset.

> Added link to the technote RTN-099: "Photometric transformation relations between \gls
> Unknown macro: {LSSTCam}
>
> and \gls
> Unknown macro: {LSSTComCam}
>
> systems and other photometric systems are under development and are provided in \citep
> Unknown macro: {RTN-099}
>
> "

Line 2269 - Applying a reliability threshold improves the purity transients but not variable stars. What reliability value is recommended for the user looking for transients. A figure may be useful to enable readers to make an informed decision on what threshold is best for their science case.

> yes added sentence describing range and column

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